Stock trading with Fuzzy RL model and Sentiment Analysis - Case Study on Big Tech Stocks

Authors Blinded

Abstract. This study proposes a reinforcement learning (RL) framework that blends LSTM-based news sentiment analysis for automatic trading in NVIDIA (NVDA). With a reinforcement learning agent that takes quantitative price information and qualitative sentiment signals as input, the model employs a sliding-window strategy: the sentiment classifier is trained on 30-day news headline windows and the RL agent is retrained every 7 days to predict the following week. Totally 980 English heading corpus from 124 sources between December 19, 2024 and March 19, 2025 were constructed, utilizing minute-by-minute closing prices for NVDA. Without sliding-window updating, the trading strategy generated a 25% return on the tests; with the proposed sliding-window retraining, it generated a +16.98% return, a result derived from 30 separate Monte Carlo simulations. These results demonstrate that updating both the sentiment model and the trading policy at a regular and brief intervals significantly improves robustness and profitability under this non-stationary market setting.

Keywords: Reinforcement Learning, Algorithmic Trading, Stock Market Prediction, Sentiment Analysis, Financial Time Series, Sliding Window, Machine Learning, LSTM, Monte Carlo Simulation, NVIDIA, TSLA, MSFT, META, GOOG, Computational Finance

1 Introduction

In modern stock exchange, profitable trading increasingly depends on the ability to fuse heterogeneous information,price micro-structure, news flow, and shifting regimes—into a single adaptive policy. Reinforcement learning (RL) is a natural fit for this sequential decision problem, yet agents often disrupt when market conditions change or when news sentiment reverses direction. To address this, we recently proposed a *hybrid* framework that combines a sliding-window machine learning forecaster for one-minute returns, and an RL policy whose state vector augments price features with sentiment scores, both retrained on a rolling schedule. While an intensive Monte-Carlo campaign on NVIDIA (NVDA) confirms the method's statistical robustness, a broader question remains:

> *Can a single parameterisation of this hybrid agent generalise across multiple large-cap names that differ in liquidity, news cadence and volatility?*

To establish a practical baseline, the model without re-tuning hyper-parameters applied the *same* six additional high-capitalisation tickers: META, INTC, MSFT, GOOGL, AAPL, and TSLA. For each stock, the agent ingested minute by minute prices and aligned news headlines between 12 December 2024 and 19 March 2025, yielding a uniform test set of 77,082 observations enriched with sentiment scores. Table 1 summarises these single-run results.

Table 1. Single-run performance of the proposed strategy versus a buy-and-hold benchmark (initial \$10 000 capital).

Ticker	Strategy Return	Buy&Hold Return	NLP Rows	Final Value (\$)	Trades
META	92.40%	−3.88%	77 082	19 239.79	542
INTC†	15.73%	30.96%	77 082	11 573.01	274
MSFT	12.66%	0.72%	77 082	11 266.29	216
GOOGL	10.99%	−1.81%	77 082	11 099.47	490
AAPL	9.17%	−3.43%	77 082	10 917.47	432
TSLA	6.18%	−1.85%	77 082	10 617.76	281

† INTC used dummy news data because the original news file was not sufficient compared to other stocks, likely affecting its relative performance.

The hybrid agent outperformed the passive benchmark on five out of seven entities.A particularly strong example of META, which came close to doubling its initial investment, is INTC the single negative outlier; its carefully crafted headlines deprive the model of genuine sentiment indicators highlighting the importance of news sentiment scores. In short, these early cross-asset findings support the functionality of the framework and motivate a deeper Monte Carlo evaluation of the Nvdia stock presented later in the article.

2 Literature Review

Sentiment analysis, reinforcement learning and machine learning algorithms are developing day by day. Deep reinforcement learning, Support Vector Machines (SVM) and Random Forests methods surpass traditional forecasting algorithms in the field of finance [1, 2, 3, 4, 5]. Fischer and Krauss used Long Short-Term Memory (LSTM) networks to improve the analysis of sequential data in the context of predicting movements within the S&P 500. Their results offer empirical support that deep learning techniques are more accurate than regression analysis in capturing the complex dynamics of financial time series data [6].

Along with the old-fashion price based forecasting schemes, natural language processing has completely changed the way qualitative information is put to use. Zeng and Jiang [7] demonstrated that using a FinBERT pre-trained language model combined with an LSTM deep network in sentiment extraction from financial news led to better prediction performance as opposed to, ARIMA, single LSTM and BERT models. The study presented by Dolphin et al. [8] proposes a framework to leverage an LLM in extracting structured company-level sentiment information from unstructured financial news.

Nandkumar and Shaik proposed a spatial federated learning framework for storing news on the blockchain, which can enhance the trustworthiness of sentiment-based stock predictions for verifiable processing [9].

Recent literature further emphasizes dynamically integrating these qualitative data streams into trading policies. Altuner and Kilimci [10] alongside Chaudhari and Mahajan [11] demonstrated that combining deep reinforcement learning and ensemble methods with community-aware sentiment analysis significantly enhances stock prediction robustness. Furthermore, Sattar et al. [12] recently introduced an RMS-driven deep reinforcement learning framework that effectively adapts to market volatility, reinforcing the value of dynamic state-space modeling in algorithmic trading.

3 Methodology

A supervised one-step-ahead return forecaster with a tabular Q-learning agent that decides whether to *buy*, *sell*, or *hold* a single share in a minute-bar environment.

3.1 Data Alignment

For each stock two synchronous streams were merged: (i) minute mid-prices P_t ($t = 1, \ldots, T$) and (ii) news headlines time-stamped in UTC. The latest headline *strictly prior* to the bar close ($t - 15\,\text{min} < \tau < t$) is converted into a sentiment score $S_t \in [0, 1]$ (Section 3.2). Missing headlines receive the neutral prior $S_t = 0.5$.

3.2 Sentiment Classification

TF–IDF logistic regressor used on a hand-labelled corpus of $\approx 10^4$ headlines.[1] The predicted posterior $\hat{S}_t = \Pr(\text{positive} \mid \text{headline})$ is clipped to $[0.01, 0.99]$ to prevent extreme rewards.

3.3 Feature Vector

Define the one-minute percentage return

$$r_{t+1} = 100\,\frac{P_{t+1} - P_t}{P_t}. \tag{1}$$

The feature vector $\mathbf{x}_t \in \mathbb{R}^{23}$ stacks *price shocks*, *volatility*, *momentum*, and *sentiment lags*:

$$\mathbf{x}_t = \left[\Delta P_{t:t-5},\, \sigma_t^{20},\, M_t^5,\, S_{t:t-5},\, \bar{S}_t^{10},\, \sigma_{S,t}^{10}\right]^\top. \tag{2}$$

[1] Training details in Appendix A.

3.4 Return Forecaster

Five different models—OLS, RF, XGBoost, MLP, and LSTM—inside a rolling window of N_{ML} days were trained. Hyper-parameters are tuned on a *pre-tail validation* slice to avoid RL leakage:

$$\boldsymbol{\theta}^{\star} = \arg\min_{\boldsymbol{\theta}} \mathrm{RMSE}_{\mathrm{val}}\big(f_{\boldsymbol{\theta}}(\mathbf{x}), r\big). \tag{3}$$

The best model outputs $\hat{r}_t = f_{\boldsymbol{\theta}^{\star}}(\mathbf{x}_t)$.

3.5 State Space

Each minute the environment emits

$$s_t = \big(b_p(\Delta P_t),\, b_s(S_t),\, b_m(\hat{r}_t),\, p_t\big), \tag{4}$$

where b_* maps its argument into $K_* = 5$ equiprobable bins and $p_t \in \{0,1\}$ flags the current long position.

3.6 Reward Function

Let P_{entry} be the last fill price and c_a the action-specific transaction cost. For $a_t \in \{\mathrm{Buy}, \mathrm{Sell}, \mathrm{Hold}\}$,

$$r_t^{\mathrm{RL}} = \begin{cases} 100\,\dfrac{P_{t+1} - P_t}{P_t} - c_{\mathrm{buy}} + \beta_m \hat{r}_t + \beta_s S_t, & \text{Buy}, \\ 100\,\dfrac{P_t - P_{\mathrm{entry}}}{P_{\mathrm{entry}}} - c_{\mathrm{sell}} - \beta_m \hat{r}_t + \beta_s (1 - S_t), & \text{Sell}, \\ 100\,\dfrac{P_{t+1} - P_t}{P_{\mathrm{entry}}} - c_{\mathrm{hold}} - \lambda h_t, & \text{Hold}. \end{cases} \tag{5}$$

3.7 Q-Learning

The Q-table updates as

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha_t \big[r_t^{\mathrm{RL}} + \gamma \max_{a'} Q_t(s_{t+1}, a') - Q_t(s_t, a_t)\big], \tag{6}$$

with $\alpha_t = 0.1$ and $\gamma = 0.95$. Exploration follows an ε-greedy schedule $\varepsilon_t = \max\big(0.01,\, 0.9 \times 0.995^t\big)$.

3.8 Rolling Training Protocol

For each window

1. Fit $f_{\boldsymbol{\theta}}^{(w)}$ on the past N_{ML} days.
2. Initialise $Q^{(w)} \leftarrow Q^{(w-1)}$ and train on the next N_{RL} days.
3. Freeze both models and evaluate on a *forward* test slice, logging terminal wealth $V_T = \mathrm{cash}_T + n_T P_T$.

Reliability is assessed via 30 Monte-Carlo runs with bootstrap resampling.

4 Findings

Using minute-bar price data in combination with newssentiment-augmented reinforcement learning (RL) agent based on sliding-window adaptation proved highly effective for high-frequency stock trading. The approach was tested on six technology stocks and managed to outperform the buy-and-hold benchmark in five instances. The only stock that faltered, INTC, was based on surrogate news data, thus highlighting the absolute importance of genuine sentiment signals. The requirement of the sliding-window component was also validated by ablation testing on NVIDIA stock, which saw an average loss of around 25% when this was removed. In addition, a Monte Carlo simulation over 30 iterations in NVDA resulted in a high mean weekly return of +14.61%, further attesting to the generalizability of the proposed approach. The number of trades placed varied from 216 (for MSFT) to 542 (for META), indicating the ability of the model to adjust to asset-specific volatility as well as its proficiency for capturing short-term trading opportunities. Overall, the results support the supposition that reinforcement learning, augmented by sliding-window adaptation and real-time sentiment analysis, forms a powerful and generalizable framework for adaptive algorithmic trading.

5 Conclusion and Future Directions

Sentiment-aware reinforcement learning approach for high-frequency trading with a sliding-window model, adaptive Q-learning, and real-time news sentiment. The experiments proved, it performs significantly better compared to baseline approaches on big-cap equities. Monte Carlo simulations and ablation studies highlighted the importance of model updates and changes in sentiment. The hybrid model has potential in unstable markets. Additional studies specified under live market conditions can advance the research by extending evaluation of the model architecture.

References

[1] Jie Zou et al. "A Novel Deep Reinforcement Learning Based Automated Stock Trading System Using Cascaded LSTM Networks". In: *arXiv preprint arXiv:2212.02721* (2022).

[2] Jigar Patel et al. "Predicting stock and stock price index movement using Trend Deterministic Data Preparation and machine learning techniques". In: *Expert Systems with Applications* 42 (Jan. 2015), pp. 259–268. DOI: `10.1016/j.eswa.2014.07.040`.

[3] Santosh Kumar Sahu, Anil Mokhade, and Neeraj Dhanraj Bokde. "An Overview of Machine Learning, Deep Learning, and Reinforcement Learning-Based Techniques in Quantitative Finance: Recent Progress and Challenges". In: *Applied Sciences* 13.3 (2023), p. 1956. DOI: `10.3390/app13031956`.

[4] Sira Yongchareon. "AI-Driven Intelligent Financial Forecasting: A Comparative Study of Advanced Deep Learning Models for Long-Term Stock Market Prediction". In: *Machine Learning and Knowledge Extraction* 7.3 (2025), p. 61. DOI: `10.3390/make7030061`.

[5] Ricardo P. Masini, Marcelo C. Medeiros, and Eduardo F. Mendes. "Machine Learning Advances for Time Series Forecasting". In: *Journal of Economic Surveys* 37.1 (2021), pp. 76–111. DOI: 10.1111/joes.12429.

[6] Thomas Fischer and Christopher Krauss. "Deep learning with long short-term memory for financial market predictions". In: *European Journal of Operational Research* 270.2 (2018), pp. 654–669. DOI: 10.1016/j.ejor.2017.11.054.

[7] Qingyun Zeng and Tingsong Jiang. "Financial sentiment analysis using FinBERT with application in predicting stock movement". In: *arXiv preprint arXiv:2306.02136* (2023).

[8] Rian Dolphin et al. "Extracting Structured Insights from Financial News: An Augmented LLM Driven Approach". In: *arXiv preprint arXiv:2407.15788* (2024). URL: https://arxiv.org/abs/2407.15788.

[9] Nitin Nandkumar Sakhare and Imambi S. Shaik. "Spatial federated learning approach for the sentiment analysis of stock news stored on blockchain". In: *Spatial Information Research* 32.1 (2024), pp. 13–27. DOI: 10.1007/s41324-023-00529-x.

[10] Anıl Berk Altuner and Zeynep Hilal Kilimci. "A novel deep reinforcement learning based stock price prediction using knowledge graph and community aware sentiments". In: *Turkish Journal of Electrical Engineering and Computer Sciences* 30.4 (2022), pp. 1506–1524. DOI: 10.55730/1300-0632.3862.

[11] Archana Y Chaudhari and Smita Mahajan. "Prediction of stock market using sentiment analysis and ensemble learning". In: *MethodsX* 14 (2025), p. 103260. DOI: 10.1016/j.mex.2025.103260.

[12] Asma Sattar et al. "A Novel RMS-Driven Deep Reinforcement Learning for Optimized Portfolio Management in Stock Trading". In: *IEEE Access* 13 (2025), pp. 42813–42835. DOI: 10.1109/ACCESS.2025.3546099.